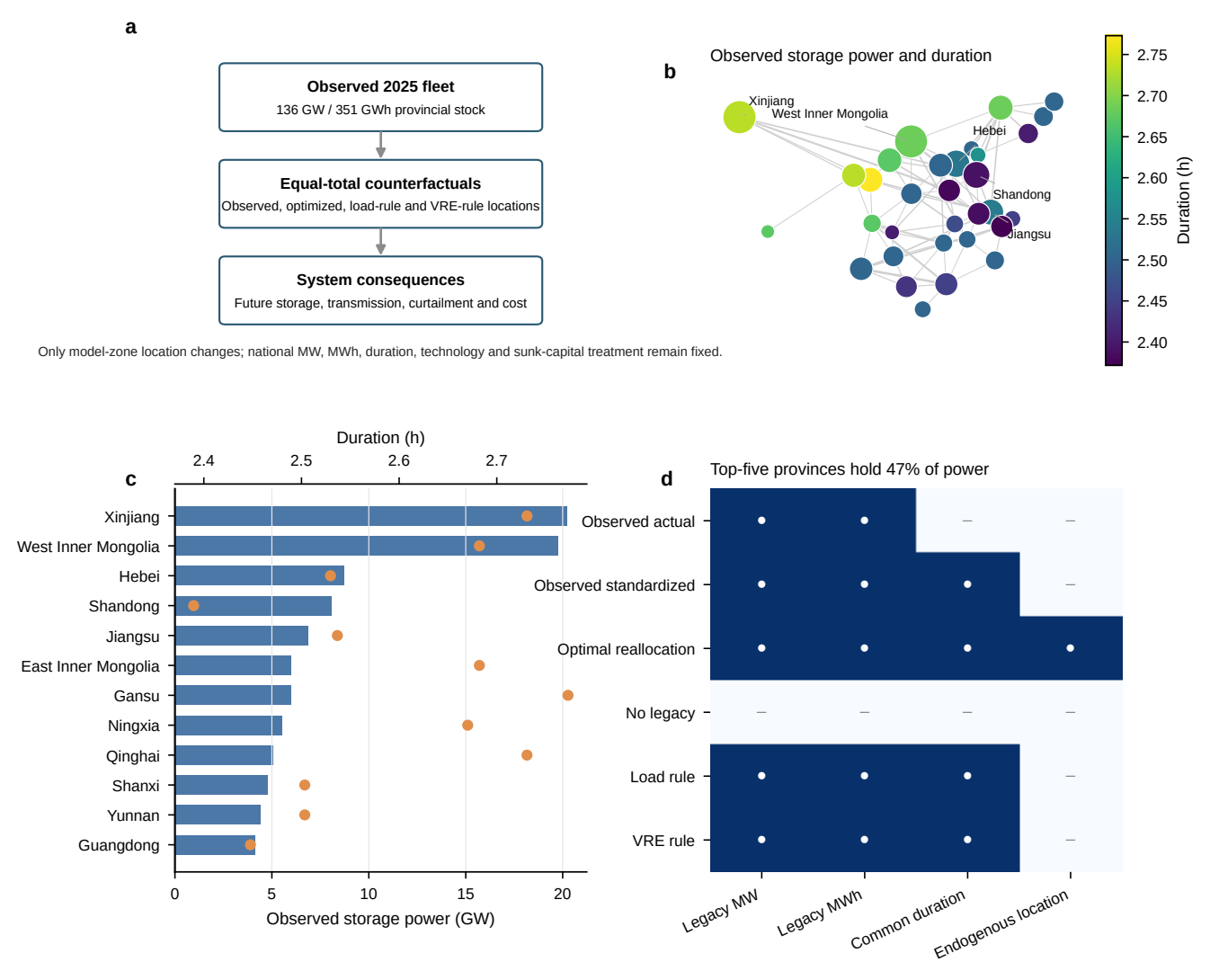
a
Observed 2025 fleet
136 GW / 351 GWh provincial stock
Equal-total counterfactuals
Observed, optimized, load-rule and VRE-rule locations
System consequences
Future storage, transmission, curtailment and cost
Only model-zone location changes; national MW, MWh, duration, technology and sunk-capital treatment remain fixed.
b
Observed storage power and duration
Xinjiang
West Inner Mongolia
Hebei
Shandong
Jiangsu
Duration (h)
2.75
2.70
2.65
2.60
2.55
2.50
2.45
2.40
c
Duration (h)
2.4
2.5
2.6
2.7
Xinjiang
West Inner Mongolia
Hebei
Shandong
Jiangsu
East Inner Mongolia
Gansu
Ningxia
Qinghai
Shanxi
Yunnan
Guangdong
0
5
10
15
20
Observed storage power (GW)
d
Top-five provinces hold 47% of power
Observed actual
Observed standardized
Optimal reallocation
No legacy
Load rule
VRE rule
Legacy MW
Legacy MWh
Common duration
Endogenous location